

# Power-Grid Structure Exploration with Unified Sequence-based Learning Framework

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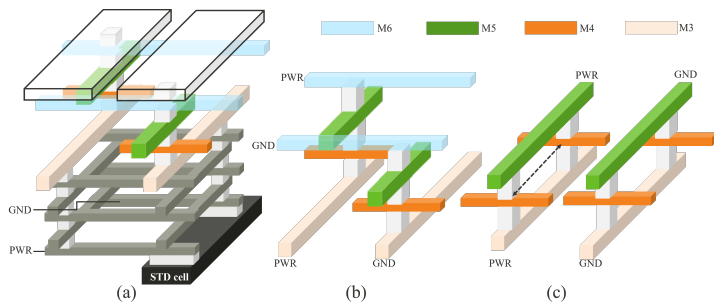
**Abstract**—As increasing gap of the shrinking ratio between device and metal pitches under advanced nodes, power-grid (PG) structure plays a critical role in circuit performance considering power integrity. Initial PG structures, as initiating implementation fundamentals, dominate optimization space. To address initial structures on industrial flow, we propose the *first* unified framework fueled by novel sequence-based structure generator and Transformer-based predictor, providing accurate static voltage-drop estimation. Learned embeddings are then adopted to determine promising candidates with Pareto frontier during structure optimization. The predictor is demonstrated in different scenarios under  $3nm$  and  $2nm$  technology with average 0.011% maximum absolute error of drop percentage while the optimized structures reduce 15% PG utilization with 34% worst timing-slack improvement on industrial designs.

## I. INTRODUCTION

As transistor size keeps shrinking, PG structures impose a crucial role in circuit performance for increasing voltage (IR) drop and reserving sufficient routability. In industrial design flow, designers carefully design a set of global PG structures with different sustainable static IR (SIR) drops to initiate physical implementation.

Since global PG design is conducted before real cell placement, SIR is usually adopted as healthiness index, and different power densities (PD) of IP designs lead to rich possibilities of PG structures. In industrial practice, designers could mix different power density structures to address local hotspots or dynamic IR drop after cell placement with much smaller design overheads.

A complete structure is defined by an architecture and corresponding pattern configuration. The architecture determines PG-style combination in stripes or pillars as illustrated in Fig. 1(a). As observed in the figure, layer architecture has various forms and the architecture hooks up PG metals among layers through proper via connections. To avoid PG occupying excessive routing resource under advanced nodes, stagger and inline pillars are the most common stub architectures, as shown in Fig. 1(b) and (c). On the other hand, the pattern configuration further details how to implement such architecture, e.g., the needed vertical spacing (pitch  $y$ ) between two VIA4 of an inline pillar in Fig. 1(c).



**Fig. 1: Power-grid structure overview. (a) Grid stacking adopted in advanced nodes with two most common pillar architectures, (b) stagger and (c) inline pillar.**

Typically IR analysis models PG as a resistor network and attaches corresponding current loads at each network node. To evaluate the quality of a global PG structure, the common practice assigns average power density simulating current loads with commercial sign-off tool. During physical implementation, finer-grain adjustment based on a global structure is made to address dynamic IR drop from local hotspots. As a result, the global PG structure has significant impacts in physical convergence.

Considering global structure optimization, it is mandatory to address the requirement of increasing and diverse power densities under deep sub-micron technology. Power density increases dramatically because of smaller logic area but much complicated circuit functions. In addition, SoC IPs have different power densities, causing unnecessary area waste or systematic IR-drop risk with single structure.

To tackle aforementioned two problems, scalable structure exploration is essential. Existing related works include two major categories, (1) IR-change prediction of different cell placements under one fixed PG [2], [4], [5], [7], [9], [11], [12], [15]–[17], and (2) PG structure adjustment with limited styles under different cell placements [1], [8], [10]. First category focuses on accurate prediction but does not address much about how to reduce IR violations. Second category aims at PG adjustment at specific current distribution strengthening local robustness; however, the explored structures are difficult to consider rich pillar combinations.

There are multiple learning models proposed in related works, including linear regression [16], XGBoost [8], [12], and U-Net [4], [5], [15]. A recent work [17] adopted tree-based model to make prediction while examining feature importance by attention calculation and pruning non-important features. Although these models are effective in static or dynamic IR prediction, we found that the model sensitivity on finer-grain structures is insufficient. Considering PG adjustment in existing works, only specific layer or style changes are made, and thus it is infeasible to be adopted in industrial cost-effective designs.

In this work, we propose scalable learning-based PG structure exploration, with SIR-driven global structure optimization, to offer sufficiently good initial solutions with various power densities reducing adjustment iterations. In industrial practice, it is less preferred changing global PG structures constantly to reset floorplan utilization. Moreover, with various density offerings, we could easily address local dynamic hotspots by mixed density structures.

Our framework addresses multi-density optimization with different structures recommended from Transformer-based learning model. To enable flexible structure exploration, sequence-based architecture perturbation is developed followed by sets of detailed pattern configuration generation. The proposed sequence-based structure operations are carefully designed to run through the model training and optimization. Compared with existing IR-drop models, this is the *first* work to formulate the structure features in transformer-based architecture, where the attention mechanism is further strengthened on filtered samples by Maximin Latin Hypercube (MLHS). Our model effectively captures inter-layer relationships over common regression models to greatly improve IR-drop prediction. Promising structure patterns are generated through non-

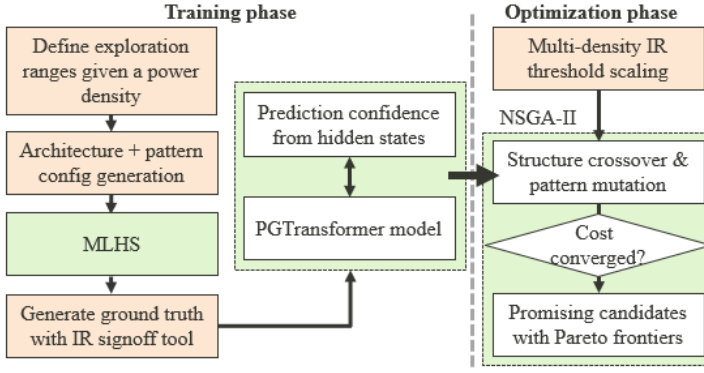


Fig. 2: Proposed framework to create optimized PG structures.

dominated sorting derived from augmented drop prediction and associated resources.

We summarize the key contributions of our proposed framework as follows:

- Compared with limited structure supports in related works, our explorer builds flexible PG pattern stacking in sequence representation to generate vast candidates followed by MLHS for representative training samples.
- To better capture subtleties of multi-layer structure changes over existing works, different pattern configurations are translated into a sequence each layer. We present PGTransformer on proposed sequence representation as core predictor to capture inter-layer relationships with self-attention mechanism.
- Proposed framework equips capability to generate optimized *multi-layer* structure patterns from Pareto frontiers. New patterns are derived from balanced trade-off between augmented drop value with prediction confidence and resource.

The rest of this paper is organized as follows. Sec. II illustrates our exploration framework and key procedures. Section III details our algorithms. Experimental results are presented in Section IV. Finally, conclusions are given in Section V.

## II. PROPOSED UNIFIED FRAMEWORK

To offer efficient structure optimization, the framework combining both training and optimization is proposed as illustrated in Fig. 2. The purpose of training phase builds a robust IR-drop prediction model to examine power-integrity quality of various structures. The optimization phase adopts the trained model along with resource and prediction confidence to explore different architecture patterns.

In model training phase, we transform multi-layer PG structures into a set of sequences for efficient perturbation and enable sequence-based model to learn structure contexts. Legal layer configurations are defined depending on target metal scheme. For each candidate layer, a valid pitch search range is defined, and we set the range in  $\{8, 16, \dots, 48\}$  times of default pitch in this work. The search range can be changed to further expand solution space or fit target technology rules.

With defined search range, our proposed structure generator creates various layer pattern stacking with pillar template combinations. The stacking operations are conducted with string manipulation and thus the generator provides highly efficient sample generation. However, to manage costly training sample collection with sign-off IR values, it is mandatory to reduce training sample sizes. After sets of structures are generated, we apply MLHS from multivariate structure parameter distribution for equidistant sample space. Commercial IR sign-off tool is then adopted to generate golden IR values of sampled training structures. The IR predictor is trained with Transformer-based architecture by forwarding sets of layer features to its embedding encoder.

As mentioned in Sec. I, it is essential to prepare structures considering different power densities for minimized area overhead. Considering IR drop mainly determined by the current multiplication, direct proportional to assigned power, IR drop is inversely proportional to structure sustaining certain power density. As a result, we expand power density supports during optimization phase by changing target drop threshold as

$$D_{th}^j = \frac{PD_i}{PD_j} \times D_{th}^i, \quad (1)$$

where  $PD_i$  is the structure sustaining power density  $i$  and  $D_{th}^i$  is the target IR-drop threshold under power density  $i$ .

In major optimization iterations, we intend to figure out most promising structures approaching  $D_{th}^j$  with best trade-off between resource and IR quality. Multi-objective optimization with Non-dominated Sorting Genetic Algorithm (NSGA-II) [6] is adopted to address the sequence-based structure representation. The optimizer identifies Pareto frontiers considering reserved routing resource, and augmented IR prediction by comparing learned embeddings.

## III. PROPOSED ALGORITHM

### A. Sequence-based Structure Generation

Proposed structure generation aims to create flexible structures among all layers efficiently. The generation includes two stages, vertical stacking to determine an architecture and pattern configuration mapping within the architecture. During generation, we define a set of discrete candidates to determine the width, number, and spacing of PG metals at each layer considering technology rules. To address metal-pitch alignment as a feasible PG, template-based permutation is proposed.

A set of pillar architecture templates is defined as illustrated in Fig. 3 and note that similar concepts can be easily extended to more complicated template designs. Each pillar template  $AT_i$  defines how a pillar is formed between two stripes under stacking layers  $i$ . To provide flexible combination, the stacking could be either bottom-up  $AT_i^{R0}$  or top-down  $AT_i^{MX}$ . Given stacking direction R0, MX, the corresponding  $x$ -axis and  $y$ -axis pitch are revised based on the preferred direction of the metal layer where template stripe lies. Pitches  $x$  and  $y$  describe cross-layer alignment with respect to an anchor layer, which could be inside a template or defined by other stacking one.

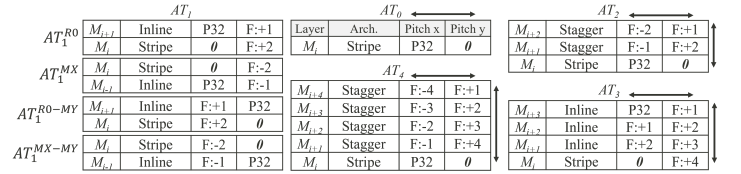
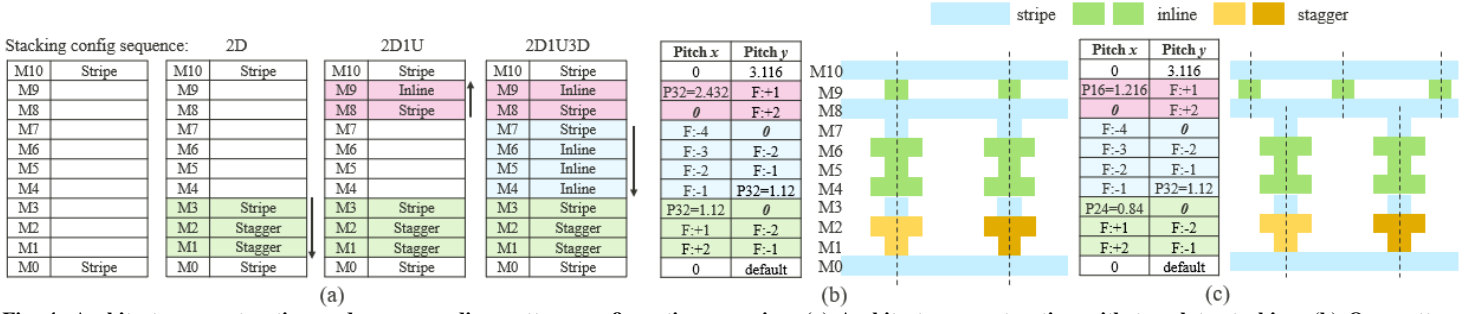


Fig. 3: Pillar template examples considering five-layer combinations.

At first stage, architectures are constructed with defined templates based on sequence permutation as depicted in Fig. 4(a). The sequences manage possible pillar template combinations in vertical stacking direction, comprising pairs of two-digit characters. The first character denotes the template index  $i$  among candidate set  $AT$  and second one describes the stacking direction, “D” for bottom and “U” for top. Every stacking needs to make template pillars enclosed by two stripes considering stacked templates or boundary conditions for a feasible pillar structure. As the example described in the figure,  $AT_2^{MX}$  is chosen and stacked from the bottom side at first operation, and similar concepts are applied to second and third operations. For third operation, although the stacking direction is from the bottom,  $AT_3^{MX}$  is chosen instead of  $AT_3^{R0}$  to enclose template pillars.

After a complete architecture is formed, detailed pattern configurations are generated at second stage. To form a valid pillar pattern, we define proper pitch relationships among related neighboring layers. As depicted



**Fig. 4: Architecture construction and corresponding pattern configuration mapping. (a) Architecture construction with template stacking. (b) One pattern configuration candidate, and (c) another configuration by changing anchor pitches under an identical architecture.**

in Fig. 4(b), certain template layers define drawing PG pitches while the pitches of other layers “follow” different anchor layers to create pillar patterns. The drawing pitch is a set of discrete numbers, described by “ $Pm_{tp}$ ” where  $m_{tp}$  denotes the multiple of default layer pitch. The follow relationship is described by a string with an important index “ $F : d_{floor}$ ”, where  $d_{floor}$  denotes the floor offset with respect to the alignment anchor layer, e.g., 0 implies current layer, 1 (-1) implies one layer above (below).

As illustrated in the example, considering vertical preferred direction, the pitch  $x$  of template string at M7 matters, and thus  $AT_3^{MX}$  is further revised to  $AT_3^{MX-MY}$  defining explicit horizontal PG pitch. Same concepts are also applied to other two templates. After template revision, by tracing follow relationships, it is clear that the pattern configuration can be manipulated through different anchor pitches, i.e.,  $1.12\mu m$  from  $32X$  M4 layer pitch  $0.035\mu m$ . By changing anchor pitches on M3 and M9 in the figure, another different patterns are formed given one identical architecture as depicted in Fig. 4(c).

Pitch configurations among all layers are then translated to sets of explicit numbers. Combined with aforementioned PG-layer width, number, and spacing, vast amounts of implementable structure configurations can be easily generated with Cadence *PSDL*, PG Structure Description Language.

### B. Feature Designs and Sampling

After pattern configuration is translated, the style and numbers at each layer are formed as a sequence of features. The features used for model training and prediction in this work are listed in Table I. The features directly derived from explicit configuration well describe a PG structure to help our predictor capture accurate inter-layer weights.

In addition to the values in pattern configuration, we also calculate  $MN_i$  and  $VN_i$  to facilitate model learning electrical behaviors because via counts are highly correlated with power-integrity robustness. Considering vertical preferred direction of Layer  $i$ , and  $S_{pillar}$  is the set containing stagger and inline styles given a floorplan dimension  $W_{fp}$  by  $H_{fp}$ ,

$$MN_i = \begin{cases} \frac{W_{fp}}{Pitch_{x,i}} \times \frac{H_{fp}}{Pitch_{y,i}} \times GN_i, & \text{if } Arch_i \in S_{pillar} \\ \frac{W_{fp}}{Pitch_{x,i}} \times GN_i, & \text{otherwise} \end{cases} \quad (2)$$

and then  $VN_i = MN_i \times \rho$ , where  $\rho = GN_i$  if either one via layer belongs to  $S_{pillar}$  or  $\rho = MN_{i+1}$ .

To manage costly training sample collection with ground truths from commercial IR sign-off tool, it is essential to have minimized dataset that adequately represents majority feature space. After a set of random pattern configurations are generated, MLHS is adopted to figure out minimized set of training samples.

MLHS begins by generating a random Latin Hypercube sample, and then iteratively improves the sample by swapping pairs of data points to increase the minimum distance between them. The swapping process continues until no further improvement can be made. Considering the

**TABLE I: Definition of important features used in power/ground nets on Layer  $i$ . Power and ground nets could have different values.**

| Notation      | Type | Description                                              |
|---------------|------|----------------------------------------------------------|
| $Arch_i$      | Cat. | Architecture type (e.g., stripe, stagger, inline)        |
| $W_i$         | Num. | Metal width of this PG net                               |
| $L_i$         | Num. | Metal length of this PG net                              |
| $GN_i$        | Num. | Metal number in one group                                |
| $GS_i$        | Num. | Spacing between metals within a group when $GN_i \geq 2$ |
| $OS_i$        | Num. | First metal offset within a group                        |
| $Pitch_{x,i}$ | Num. | Horizontal pitch between metal groups                    |
| $Pitch_{y,i}$ | Num. | Vertical pitch between metal groups                      |
| $R_i$         | Num. | Resistance of Layer $i$                                  |
| $MN_i$        | Num. | Total metal counts of this PG net                        |
| $VN_i$        | Num. | Total via counts of this PG net                          |

features comprising both numerical and categorical types, we measure their distances differently. The pairwise distance between two samples  $s_i$  and  $s_j$ , is defined as follows,

$$d_\lambda(s_i, s_j) = \sqrt{\sum_n |s_i^n - s_j^n|^2 + (1 - \lambda) \times \sum_c |s_i^c - s_j^c|} \quad (3)$$

where  $s_i^n$  and  $s_i^c$  represents the numerical and categorical features, respectively, and  $\lambda$  is the weight to balance these two types. We intend to find the minimal distance from sample set  $\mathcal{S}$  with total  $N$  features,

$$d_\lambda(\mathcal{S}) = \min(\{0 \leq i < j \leq N : d_\lambda(s_i, s_j)\}) \quad (4)$$

Different  $\lambda$  in a discrete step ranged in  $[0, 1]$  are swept with random generated sample candidates, and the sample set with the maximal  $d_\lambda(\mathcal{S})$  among iterations is chosen as final training set.

### C. PGTransformer Architecture

As described in Sec. III-B, each layer of a PG structure is characterized by distinct features such as pitch, spacing, and width. We formulate layer features into a sequence with length  $L$  and trained on Transformer-based model [14] for its superior capability of capturing temporal dependencies in sequence data. Specifically, the sequence representing a structure with  $L$  layers is defined as  $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_L\}$  where  $\mathbf{x}_i = \{Arch_i, W_i, \dots\}$ .

Transformer architecture comprises  $N$  blocks, each containing a multi-head attention layer, a fully connected feed-forward network, and two normalization and residual addition layers. The Transformer can be viewed as a function  $TF(\cdot) : \mathbb{R}^{L \times d} \rightarrow \mathbb{R}^{L \times d}$  that accepts a sequence of length  $L$  of input tokens  $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_L\}$ ,  $\mathbf{x}_i \in \mathbb{R}^d$  as input, and outputs a sequence of token embeddings  $\hat{\mathbf{X}} = \{\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \dots, \hat{\mathbf{x}}_L\}$ ,  $\hat{\mathbf{x}}_i \in \mathbb{R}^d$ , where each  $\hat{\mathbf{x}}_i$  encodes the dependencies between  $\mathbf{x}_i$  and other input tokens.

The detailed structure of the Transformer encoder is depicted in Fig. 5(a). Positional embeddings are added to  $\mathbf{X}$  to encode the position information of each token. The multi-head self-attention layer with total  $H$  heads, learns independent transformations  $\mathbf{W}_h^Q$ ,  $\mathbf{W}_h^K$ , and  $\mathbf{W}_h^V \in \mathbb{R}^{d \times d}$  at the  $h$ -th head and calculates the attention weights  $\mathbf{A}_h$  among all input tokens. Token embeddings are updated with the context information

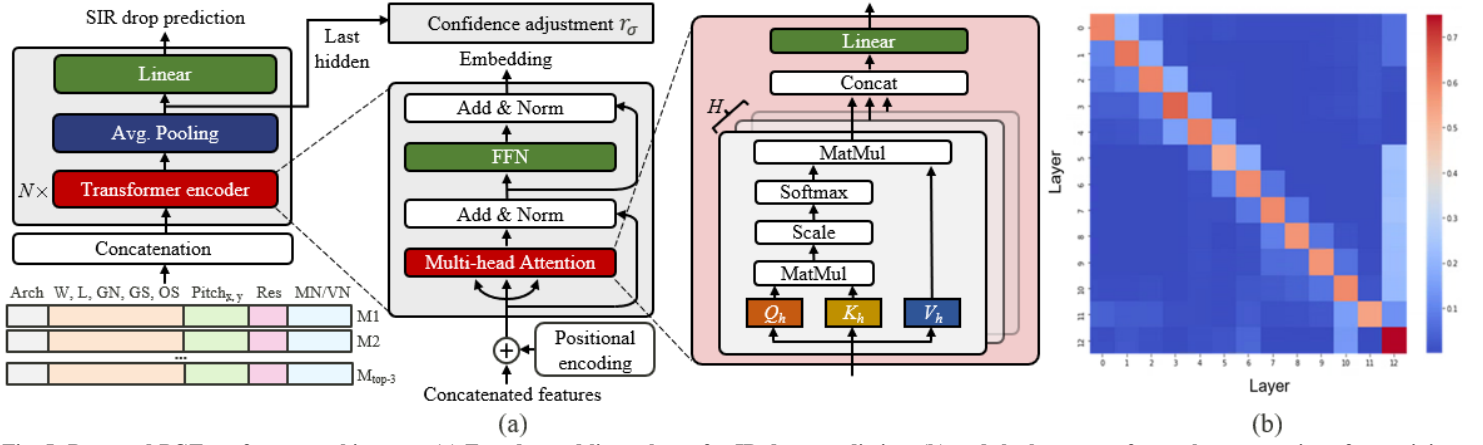


Fig. 5: Proposed PGTransformer architecture. (a) Encoder and linear layer for IR-drop prediction, (b) and the heatmap of cross-layer attention after training.

from their attention tokens, and resulting embeddings from all heads are concatenated and projected to the input dimension via learnable matrix  $\mathbf{W}^O \in \mathbb{R}^{(H \times d) \times d}$ . For each  $x_i \in \mathbf{X}$ , the output embedding  $\hat{x}_i$  through the multi-head attention, can be derived as follows,

$$\mathbf{A}_h(x_i, x_j) = \text{softmax} \left( \frac{\mathbf{W}_h^Q \mathbf{X} \cdot (\mathbf{W}_h^K \mathbf{X})^T}{\sqrt{d}} \right) \quad (5)$$

$$\hat{\mathbf{X}} = \text{concat}_{h=1}^H \left\{ \sum_{j=1}^N \mathbf{A}_h(x_i, x_j) \mathbf{W}_h^V \mathbf{X} \right\} \cdot \mathbf{W}^O \quad (6)$$

Output embedding  $\hat{\mathbf{X}}$  is then passed through a feed-forward network (FFN) for non-linear mapping, allowing the model to capture complex dependencies of different categorical and numerical features in a input sequence. For output layer, our model applies AvgPooling on learned embeddings transforming  $\mathbb{R}^{N \times d}$  to  $\mathbb{R}^{1 \times d}$ , and subsequently followed by a linear layer projecting to final IR prediction value  $\in \mathbb{R}^1$ . In this work, the embedding dimension is 64 and FFN dimension is 256 with  $N = 2$  and  $H = 8$ . The encoder produces hidden states with a dimensionality of 64, which are further used to adjust prediction values by proposed quadratic error regressor in Sec. III-D.

The heatmap shown in Fig. 5(b) represents the attention map of internal attention states. The  $i$ -th row corresponds to the attention map to predict the IR value of Layer  $i$ . In addition, it is clear that the model also refers to states from all layers to form its prediction. The layer influences are adaptively adjusted through our sequence modelling.

#### D. Structure Optimization

Multi-objective Pareto frontier is adopted to create optimized structures. To reach best design performance, not only IR robustness but left signal routing resource needs to be considered. To maximize left resource, the PG utilization should be minimized. Two objectives, PG resource utilization  $c_{res}$  and augmented IR-drop prediction  $c'_{ir}$  incorporating prediction value  $c_{ir}$  and associated confidence, are formulated in our structure optimizer. The PG utilization estimation considering architecture is defined as,

$$c_{res} = (W_{fp} \times H_{fp})^{-1} \sum_i^L MN_i \times A_i \times TP_i, \quad (7)$$

where  $A_i = L_i \times W_i$  and  $TP_i$  is default pitch divided by default width of Layer  $i$ .

As PG structures have vast variants and thus not every prediction has identical accuracy, the prediction confidence measured from model hidden states is proposed to provide another perspective during structure cost evaluation. As the last layer of PGTransformer is a linear layer, the mapping from the hidden state to final prediction is also linear mapping. If the hidden state is far from the distribution of training data, it implies

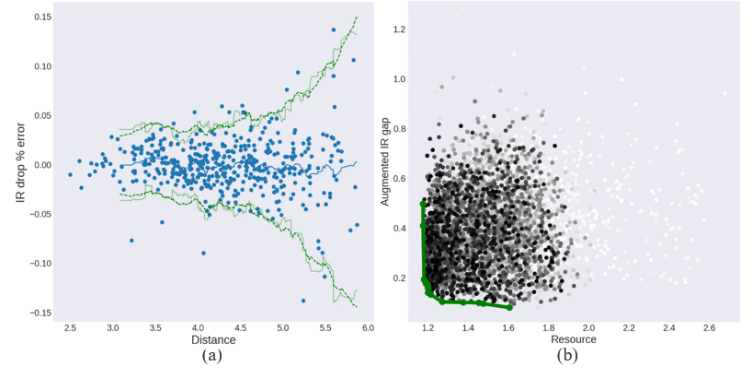


Fig. 6: Confidence cost quadratic regression and Pareto frontiers of structure optimization. (a) Rolling error mean curves with window 40 regarding  $d(x_i, S)$ , and dark green curves are proposed quadratic polynomial regressor. (b) Optimized structure frontiers after iterations.

that the prediction could be extrapolated, leading to unreliable results. We approximate prediction confidence by fitting  $\sigma$ -curve of hidden-state distance during model validation.

For each validation sample  $x_i$ , the minimum distance of its hidden states toward training dataset  $S$  as  $d(x_i, S) = \min\{\|x_i - s_j\|\}$  and its associated prediction error are measured. As shown in Fig. 6(a), the validation error along with different distances is illustrated, where  $x$ -axis is  $d(x_i, S)$ . The blue dots ( $y$ -axis) represent validation errors of IR-drop percentage,  $V_{ir}/V_{ideal}$  where  $V_{ideal}$  is supply voltage and  $V_{ir}$  is the drop value. To capture proper confidence penalty, the corresponding rolling mean and standard deviation  $\sigma$  are examined. The blue curve denotes corresponding rolling mean  $\mu$  and light green curves are  $\mu \pm 2\sigma$  from validation set. By observing  $\mu \pm 2\sigma$  curves, we adopt quadratic regression to determine the parameters as confidence cost function.

As observed in the figure, the  $2\sigma$  quadratic regression as confidence evaluation is positively correlated with both distance and original  $\mu \pm 2\sigma$  distribution with 95% sample coverage, reaching correlation coefficient 0.82 and R-squared value 0.92, respectively. To address the confidence directly into predicted IR toward a defined drop threshold  $D_{th}$ , we define the IR-drop objective in our structure optimizer as

$$c'_{ir} = |c_{ir} - D_{th}| + r_\sigma(d(x_i, S)), \quad (8)$$

where  $r_\sigma$  is  $2\sigma$  quadratic regressor of minimum state distance.

With  $c_{res}$  and  $c'_{ir}$ , we employ modified NSGA-II as multi-objective optimizer. NSGA-II works by generating a population of candidate solutions and evolving population over generations. The architecture sequence is used to form chromosomes, and corresponding pillar templates provide

good crossover points. The crossover points offer strategic perturbation, potentially combining both parent strengths.

Fig. 7 illustrates the configuration evolution process. The pairs of template stacking with equal length from each configuration pair are collected as crossover candidates. One of the pairs is randomly selected to produce new configurations. After architecture crossover, the valid pitches  $x$  and  $y$  are examined based on the preferred direction of template stripe layers, as described in bold blue in the example. In addition, the anchor pitches of two child configurations are changed with mutation probability  $n_{AT}^{-1}$ , where  $n_{AT}$  is total template number of parent configurations. With aforementioned operations, structure configurations are gradually optimized.

After the optimization converges, we determine a set of “non-dominance” candidates with Pareto frontiers. In the context of multi-objective optimization, these frontiers represent the set of all Pareto optimal solutions. Considering  $c_{res}$  and  $c'_{ir}$  as our objectives, a structure  $x^*$  is considered Pareto optimal if there does not exist another solution  $x$  such that  $c_{res}(x) \leq c_{res}(x^*)$  and  $c'_{ir}(x) \leq c'_{ir}(x^*)$  for both objectives and  $(c_{res}(x^*), c'_{ir}(x^*))$  is strictly better than at least one objective than other structures.

As depicted in Fig. 6(b) with two axes  $c_{res}$  and  $c'_{ir}$ , scattering points represent gradual converging structure candidates toward  $D_{th}$ , and iterations are illustrated in white toward black dots. Among these candidates, the frontiers are identified in green lines to provide opportunities to best fit different design characteristics.

#### IV. EXPERIMENTAL RESULTS

The prediction accuracy of PGTransformer is examined compared with several prominent models, and then optimization results empowered by the model sustaining different power densities are discussed.

##### A. Model Accuracy Comparison

To validate flexibility of the proposed framework, we prepared five sets of data in different cell heights, top-metal mesh designs, and metal schemes in TSMC N3E and N2 process nodes. The designs are implemented in flip-chip package with  $155\mu m$  bump pitch and supply voltage is set to  $0.75V$ . Separate models are trained for each of these two nodes.

Proposed structure generator randomly generates 12000 structure samples, and 3600 samples are selected by MLHS at each set where ground truths are generated by sign-off tool Cadence *Voltus*. For one set, the pool generation and sampling are done in 10 minutes from efficient string manipulation depicted in Sec. III-A, and ground truth collection takes 118 hours with distributed 2500 machine CPUs. The dataset uses 3:1 to partition into training and validation sets, respectively.

The model is implemented in PyTorch and trained on NVIDIA RTX 3090 with batch size of 16. We employ Lion [3] as the optimizer and set the weight decay to  $1e^{-2}$ . To train PGTransformer, we adopt an initial learning rate of 0.0001 with a cyclic learning rate schedule [13] and training epoch is set to 2500. For model comparison, three model architectures in related works, U-Net, XGBoost, and LSTM are also implemented.

To properly utilize U-Net, PG patterns and target PD are converted to tile-based map as [5], and we employ a bi-directional LSTM to align with the two-layer encoder in PGTransformer. For XGBoost, we use linear booster, max depth 10, subsample 1, eta 0.1, and number of boost round 100 with same learning rate. The structure generator and Pareto frontier are implemented in TCL and Python with 24-core Intel Xeon CPUs at 2.7GHz, respectively.

Table. II shows the validation error comparison of predicted IR-drop percentages. Column “ $M_{top}$ ” describes the top-mesh PG in either mesh or mixed with signal routes. Five metrics are used to measure the quality, mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and Pearson correlation coefficient (CC) and passing rate (PR). The PR is defined as the percentage among

entire validation set where the gap between predicted drop percentage and ground truth lies in defined constraint (0.1%, 0.05%, 0.03% over ideal supply voltage). Column “IT” describes model inference time on one structure.

It is clear that PGTransformer achieves smallest errors among different data scenarios. Through the sequence formulation, we further strengthen PGTransformer capturing temporal dependencies in variable-length layer features. This trait is particularly useful to understand inter-layer IR impacts with subtle pattern changes over XGBoost and U-Net, and LSTM with potential vanishing gradient problem.

In addition to the smallest loss compared with other models, Fig. 8 also shows better convergence during validation. As observed, PGTransformer consistently reaches converged and smallest loss after 500 epoch; however, XGBoost and LSTM even have worse loss before 1000 (500) epoch than U-Net in N2 (N3E).

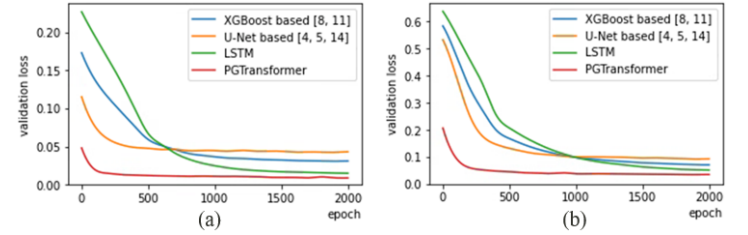


Fig. 8: Comparison of validation loss convergence between PGTransformer and prominent models in, (a) N3E 143H under 14M mixed  $M_{top}$ , and (b) N2 H130 under 16M mesh  $M_{top}$ .

##### B. Comparison with Well-Optimized Manual PG

To examine the effectiveness of structure optimization, the optimization framework is adopted on industrial designs compared with well-optimized manual works used in a N3E production and N2 test chip. The design statistics and target power density are summarized at left-hand side of Table III including high-speed CPU and power-efficient SoC IPs. Column “Period (ns)” denotes the target clock period, “CH” denotes adopted cell height, “PD” denotes target power density ( $mW/mm^2$ ), “UR (%)” denotes initial floorplan utilization setup, “#Gates ( $10^6$ )” and “#Nets ( $10^6$ )” are total instance and net counts respectively.

We set base N3E IR threshold to 1.9% with PD 6000 and N2 1.7% over ideal supply voltage to align the objective of manual PG, and corresponding IR thresholds of other densities are scaled following Eqn. 1. Physical implementation with PGs are initiated with commercial tool Cadence *Innovus* in identical flow setup and adopt Synopsys *PrimeTime PX* to report power.

The right-hand side of Table III illustrates the circuit performance (QoR for short) comparison between manual PG and the optimized one achieving smallest WNS from Pareto frontiers. Column “ $V_{ir}(\%)$ ” denotes reported drop percentage from *Voltus*, “ $D_{mx}(\%)$  ( $D_{my}$ )” denotes total PG metal density lower (above) M5, “Area ( $\mu m^2$ )” denotes post-route cell area, DRC is routing violation count, and “Opt time (min)” depicts the time reaching an optimized PG. As shown in the table, after the optimization, a new structure is created with comparable IR-drop margin while releasing more routing resource and better timing. The framework also offers great efficiency improvement to facilitate global PG designs, where the Pareto optimization with presented sequence manipulation takes average 65 minutes to converge over 2–3 week manual efforts.

Through our optimization empowered by PGTransformer, the sign-off IR quality is closer toward the defined threshold for better trade-off between threshold constraint and resource utilization. For design 5, we even identify a solution satisfying the threshold than manual one. The generated structures with averagely comparable IR-drop margins use less PG resources of lower layers by 10% and upper layers by 21%, leading to better pin access and timing optimization. With more released routing

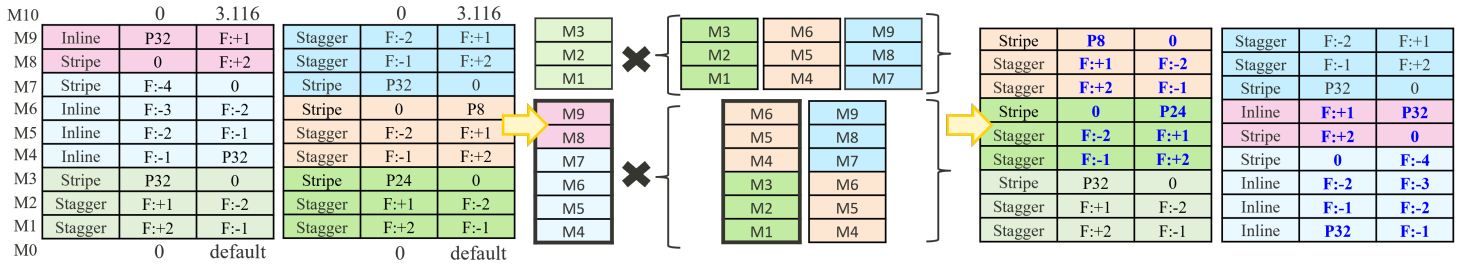


Fig. 7: Key structure crossover procedures used in modified NSGA-II to produce next generation.

TABLE II: Prediction accuracy comparison between PGTransformer and state-of-the-art models, showing PGTransformer better captures subtle structure changes. Note that it is inapplicable to compare exact models in previous works because the needed feature set and assumption (e.g., features only exist in dynamic IR analysis) are different. For fair comparison, key model architectures in related works are implemented with identical feature source.

| Cell height | Scheme | $M_{top}$ | Model                      | MAE    | RMSE   | MAPE   | CC     | $PR \leq 0.1\%$ | $PR \leq 0.05\%$ | $PR \leq 0.03\%$ | IT (sec) |
|-------------|--------|-----------|----------------------------|--------|--------|--------|--------|-----------------|------------------|------------------|----------|
| N3E 143H    | 13M    | Mixed     | U-Net based [4], [5], [15] | 0.0142 | 0.0179 | 0.0075 | 0.9348 | 1.0000          | 0.9840           | 0.9280           | 0.0013   |
|             |        |           | XGBoost based [8], [12]    | 0.0168 | 0.0207 | 0.0089 | 0.9304 | 0.7040          | 0.3360           | 0.1840           | 0.0142   |
|             |        |           | LSTM                       | 0.0091 | 0.0125 | 0.0048 | 0.9976 | 1.0000          | 1.0000           | 0.9680           | 0.0061   |
|             |        |           | <b>PGTransformer</b>       | 0.0060 | 0.0100 | 0.0032 | 0.9985 | 1.0000          | 1.0000           | 0.9838           | 0.0035   |
|             |        |           |                            |        |        |        |        |                 |                  |                  |          |
| N3E 195H    | 13M    | Mesh      | U-Net based [4], [5], [15] | 0.0059 | 0.0090 | 0.0058 | 0.8758 | 0.8645          | 0.8294           | 0.7824           | 0.0011   |
|             |        |           | XGBoost based [8], [12]    | 0.0052 | 0.0080 | 0.0045 | 0.9189 | 0.9669          | 0.8430           | 0.5207           | 0.0105   |
|             |        |           | LSTM                       | 0.0028 | 0.0042 | 0.0024 | 0.9983 | 1.0000          | 1.0000           | 1.0000           | 0.0021   |
|             |        |           | <b>PGTransformer</b>       | 0.0023 | 0.0040 | 0.0023 | 0.9988 | 1.0000          | 1.0000           | 1.0000           | 0.0034   |
|             |        |           |                            |        |        |        |        |                 |                  |                  |          |
| N3E 143H    | 14M    | Mixed     | U-Net based [4], [5], [15] | 0.0426 | 0.0575 | 0.0119 | 0.8948 | 0.5351          | 0.2813           | 0.1120           | 0.0013   |
|             |        |           | XGBoost based [8], [12]    | 0.0309 | 0.0415 | 0.0096 | 0.9077 | 0.5932          | 0.3051           | 0.2203           | 0.0112   |
|             |        |           | LSTM                       | 0.0151 | 0.0208 | 0.0047 | 0.9951 | 1.0000          | 0.9661           | 0.8644           | 0.0043   |
|             |        |           | <b>PGTransformer</b>       | 0.0088 | 0.0106 | 0.0034 | 0.9973 | 1.0000          | 0.9792           | 0.9153           | 0.0083   |
|             |        |           |                            |        |        |        |        |                 |                  |                  |          |
| N3E 143H    | 14M    | Mesh      | U-Net based [4], [5], [15] | 0.0107 | 0.0261 | 0.0089 | 0.8226 | 0.8589          | 0.7834           | 0.4829           | 0.0014   |
|             |        |           | XGBoost based [8], [12]    | 0.0069 | 0.0155 | 0.0053 | 0.8975 | 1.0000          | 0.8333           | 0.6500           | 0.0131   |
|             |        |           | LSTM                       | 0.0035 | 0.0072 | 0.0025 | 0.9929 | 1.0000          | 0.9234           | 0.8838           | 0.0026   |
|             |        |           | <b>PGTransformer</b>       | 0.0030 | 0.0055 | 0.0022 | 0.9956 | 1.0000          | 0.9765           | 0.9234           | 0.0039   |
|             |        |           |                            |        |        |        |        |                 |                  |                  |          |
| N2 H130     | 16M    | Mesh      | U-Net based [4], [5], [15] | 0.0931 | 0.1223 | 0.1821 | 0.9596 | 0.6621          | 0.3841           | 0.3170           | 0.0013   |
|             |        |           | XGBoost based [8], [12]    | 0.0720 | 0.0972 | 0.1130 | 0.9765 | 0.7330          | 0.5417           | 0.3192           | 0.0149   |
|             |        |           | LSTM                       | 0.0526 | 0.0554 | 0.0240 | 0.9923 | 0.9352          | 0.6801           | 0.4544           | 0.0045   |
|             |        |           | <b>PGTransformer</b>       | 0.0360 | 0.0517 | 0.0217 | 0.9962 | 0.9411          | 0.7738           | 0.5199           | 0.0082   |
|             |        |           |                            |        |        |        |        |                 |                  |                  |          |

TABLE III: QoR comparison between ours and manual PG structures adopted in real tapeout.

| Circuit           | Node | Period | CH  | PD   | UR | #Gates | #Nets | PG     | $V_{ir}$ | $D_{mx}$ | $D_{my}$ | WNS    | TNS     | Area  | DRC | Opt time    |
|-------------------|------|--------|-----|------|----|--------|-------|--------|----------|----------|----------|--------|---------|-------|-----|-------------|
| design 1          | N3E  | 1.111  | 143 | 3500 | 71 | 1.22   | 1.53  | Manual | 1.7      | 35.45    | 98.06    | -0.029 | -5.983  | 84339 | 20  | $\geq 2400$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.84     | 30.71    | 71.22    | -0.014 | -2.942  | 84425 | 53  | 30.42       |
| design 2          | N3E  | 0.714  | 143 | 4000 | 64 | 0.38   | 0.41  | Manual | 1.87     | 35.19    | 115.09   | -0.016 | -3.490  | 22648 | 127 | $\geq 2400$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.87     | 37.52    | 81.18    | -0.005 | -1.400  | 22486 | 132 | 33.33       |
| design 3          | N3E  | 1.429  | 143 | 5500 | 70 | 0.74   | 0.87  | Manual | 1.9      | 53.86    | 107.27   | -0.002 | -0.510  | 45760 | 186 | $\geq 4800$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.9      | 50.02    | 86.95    | -0.002 | -0.399  | 45466 | 102 | 62.38       |
| design 4          | N3E  | 0.307  | 195 | 3000 | 64 | 1.23   | 1.26  | Manual | 1.93     | 23.55    | 60.23    | -0.006 | -1.204  | 78741 | 152 | $\geq 2400$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.85     | 21.39    | 61.41    | -0.004 | -0.977  | 78928 | 158 | 13.72       |
| design 5          | N3E  | 0.290  | 195 | 4500 | 59 | 1.16   | 1.20  | Manual | 1.99     | 39.00    | 78.14    | -0.014 | -8.135  | 76129 | 199 | $\geq 4800$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.87     | 23.75    | 63.96    | -0.013 | -7.145  | 75318 | 128 | 60.78       |
| design 6          | N3E  | 0.307  | 195 | 6000 | 51 | 1.34   | 1.37  | Manual | 1.87     | 83.21    | 111.44   | -0.014 | -7.551  | 85177 | 337 | $\geq 7200$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.9      | 80.89    | 102.29   | -0.009 | -4.164  | 83998 | 256 | 92.78       |
| design 7          | N2   | 0.891  | 130 | 7500 | 65 | 0.35   | 0.42  | Manual | 1.64     | 32.57    | 137.79   | -0.053 | -19.526 | 21280 | 183 | $\geq 7200$ |
|                   |      |        |     |      |    |        |       | Ours   | 1.68     | 31.95    | 77.28    | -0.033 | -15.924 | 21330 | 188 | 163.63      |
| Avg reduction (%) |      |        |     |      |    |        |       |        | -        | -9.5%    | -20.6%   | -33.5% | -32.4%  | -0.5% | -   | -           |

resources, cell proximity has better chances to be closer and thus enables *Innovus* to conduct easier timing-slack improvement by 34%. In addition, the resource advantage further reduces usages of large driving cells with 0.5% smaller cell area.

## V. CONCLUSIONS

We have proposed a unified framework to predict SIR drop and optimize global PG structures, with novel sequence-based structure representation. Through MLHS and direct pattern modeling into PGTransformer, the prediction well captures multi-layer interactions. In industrial designs, our model outperforms state-of-the-art models by average 51% MAE reduction and 29% error improvement under small tolerance. In addition, generated structures from Pareto-frontier optimization achieves

15% smaller PG utilization and 34% timing-slack improvement on average under defined drop threshold.

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